

SCEPTRE

a recursive separator-tree detector for two-area load frequency control under stealthy false-data injection

THE PROBLEM

An attack built as $a = Hc$ leaves the weighted-least-squares residual $r = P_{\perp}z$ exactly unchanged, because $P_{\perp}H = 0$ identically.

0%

of null-space attacks are caught by the χ^2 bad-data test -- at EVERY severity

and it is cheap:

8.3 of 304 meters -- 2.7% of IEEE 118-bus -- buy a perfectly stealthy attack

and the standard defence cannot see it:

a 2-layer message-passing model reaches only 9.1% of IEEE 118-bus. No amount of training widens a receptive field.

(all three measured, not asserted -- Figs. 4, 6, 13)

THE MECHANISM

1. NESTED DISSECTION

recursively split the grid on its own vertex separators (George, 1973)

2. RSTE encoder

one weight-tied bottom-up merge, then $K = 2$ top-down broadcast sweeps

3. HALO localization

descend that same tree with conformal p-values + Benjamini-Hochberg per family

4. CBF shield

project the AGC command onto the conformally tightened safe set

the tree is built ONCE from the topology, and every weight is shared across its levels

WHAT IT BUYS

0.667

detection F1 on IEEE 118-bus, zero-shot from a 14-bus model

identical

weight tensors at $N = 14$ and $N = 118$ (1,447,111 either way). 6 of 10 baselines: undefined here

6.9x

fewer localization probes than the flat per-bus test (17.2 vs 118)

-0.0013

detection F1 lost by a random tree of identical depth and parameter count

3 seeds, paired t-tests, decision threshold frozen across strata. The IEEE 14-bus aggregate is NOT claimed as a win -- see Fig. 8.